

CHETHANA GIMHAN

No. 330/B, William Jayasuriya M/W, Kesbewa, Piliyandala

☎ +94 71 656 6257 | ✉ chethanag.23@cse.mrt.ac.lk |  [LinkedIn](#) |  [GitHub](#) |  [Portfolio](#)

OVERVIEW

Third-year Computer Science & Engineering undergraduate at the University of Moratuwa, specializing in Cyber Security. Experienced in building and deploying containerized DevOps pipelines, database design, and database forensics. Passionate about cyber security areas such as penetration testing, application security, and secure system design.

EDUCATION

University of Moratuwa

March 2024 - Present

BSc Engineering Honours in Computer Science & Engineering

Specialization: Cyber Security

CGPA: 3.55/4.0

Expected Graduation: 2028

Royal College - Colombo

2014 - 2022

G.C.E. A/L 2022 - Physical Science Stream

3 A's with Z-score 2.1068

PROJECTS

AgriSenseNet (CropWise) – Smart Agriculture Platform

2026

Large-Scale Software Engineering Group Project – Docker, GitHub Actions, GHCR, ArgoCD

- Containerized 14 microservices — FastAPI backend, React dashboard, MQTT-Kafka bridge, Spark, and 6 ML inference services — authoring 27 Dockerfiles with multi-stage builds
- Built GitHub Actions CI with Buildx caching and SHA-tagged images to GHCR, bridging to GitOps via repository dispatch to auto-update Kubernetes manifests for ArgoCD
- github.com/AgriSenseNet

Kandypack – Rail & Road Supply Chain System

2025

Database Systems Group Project – PostgreSQL, Elysia.js, Next.js, Docker

- Designed the relational schema and ERD for a 15+ entity logistics platform (orders, train/truck trips, shipments, worker rosters) and contributed to enforcing business rules — driver working-hour limits, train-capacity overflow — via triggers, procedures, and views
- Built backend API endpoints on Elysia.js (Bun runtime) with JWT authentication, and implemented the Store Manager Dashboard on Next.js
- github.com/sahithyandev/kandypack-system

Nano Processor

2025

Hardware Design Project – Xilinx Vivado, VHDL, Basys 3 FPGA

- Designed and implemented a custom 8-register processor from scratch, extending a 4-bit base core into a 5-bit datapath with a dedicated ALU and a hardware multiplier
- Built the instruction decoder for a 13-bit instruction set supporting immediate load, negation, conditional register-compare jumps, and ALU operations, plus a 7-segment display driver for on-board FPGA output
- github.com/ChethanaGimhan/nano-processor

FactumDB – MySQL Forensic Correlation Tool (Ongoing)

2026

Semester 5 Cyber Security Group Project – Python, SQLite, MySQL InnoDB Internals

- Built the utility adapter layer (`ibd2sdi`, `innochecksum`, `ibd2sql`, `mysqlbinlog`) parsing raw forensic tool output into canonical models, including positional-field-to-schema mapping and table-level filtering for multi-table binary logs
- Designed and implemented the SQLite persistence layer — schemas, physical records, binlog events, and utility-execution provenance — with normalized test fixtures validating each adapter
- github.com/Factum-Labs

HR Interview Automation System (Ongoing)

2026

Independent Project – FastAPI, PostgreSQL, Docker, GitHub Actions, Trivy, GHCR

- Built a FastAPI + PostgreSQL backend with Pydantic validation, API-key authentication, rate limiting, and structured logging, containerized with multi-stage Docker builds
- Owned the delivery pipeline — GitHub Actions CI running pytest and Trivy image scanning on every build, versioned images published to GHCR, deployed live on Render
- github.com/ChethanaGimhan/hr-interview-system

SKILLS

Languages Python, C/C++, Java, JavaScript, SQL, VHDL

Backend Python (FastAPI), JavaScript (Node.js)

Frontend React.js, Next.js, Flutter, HTML, Tailwind CSS

Database PostgreSQL, MySQL, ER Modeling, Schema Design & Normalization

DevOps & Cloud Docker, Git, GitHub Actions (CI/CD), GHCR, ArgoCD, Kubernetes, Terraform

Systems & Other Linux, Trivy, Networking, VHDL/FPGA design

LEADERSHIP & EXTRA-CURRICULAR ACTIVITIES

Vice-captain, Men's Athletic Team, University of Moratuwa

Sep 2025 - Present

Competitive 400m hurdler; supports team coordination and training alongside the team captain.

President, Mora Athletics Society

Jan 2026 - Present

Leads committee planning and coordination for university athletics events, including Mora Kid Athlete — an island-wide junior athletic meet.

Events Committee Co-lead, IEEE CIS Chapter, University of Moratuwa

Apr 2026 - Present

Co-leads event planning and coordination for the IEEE Computational Intelligence Society student chapter.

Program Committee, Hit the Grounds

Jan 2026

Served on the program committee, contributing to planning, coordination, and delivery of the event.

Student Member, Institution of Engineers, Sri Lanka (IESL)

Mar 2025 - Present

Student membership of the national professional body for engineers in Sri Lanka.

ACHIEVEMENTS

Semi-Finalist, SLIoT Challenge 2025 (Open Category)

SLT Mobitel, IESL, Department of CSE, University of Moratuwa

University Colours, Athletics

University of Moratuwa

Bronze Medal, uMora Maths Competition 2020

ENTC Department, University of Moratuwa

CERTIFICATIONS

AWS Academy Graduate – Cloud Foundations

Feb 2026

AWS Academy Graduate – Microservices and CI/CD Pipeline Builder

Feb 2026

REFERENCES

Prof. Dulani Meedeniya

Senior Lecturer

Department of Computer Science & Engineering

University of Moratuwa, Sri Lanka.

✉ dulanim@cse.mrt.ac.lk

Dr. Adeesha Wijayasiri

Senior Lecturer

Department of Computer Science & Engineering

University of Moratuwa, Sri Lanka.

✉ adeeshaw@cse.mrt.ac.lk